

Generative, Sculptable 3D Towns from Symbolic Input

Progress report - June 2026 | Danvi Simhadri | GenerativeTowns / SDFusion

1. Goal & novelty

Objective. Turn building footprints (from an OpenStreetMap tile or an uploaded footprint image) into a navigable 3D town in which each building is generated from purely symbolic input - (footprint polygon, class, height, style), with NO reference image - and is interactively sculptable: the user edits a building and it adapts into a coherent structure (in the spirit of real-time SDF tools such as Unbound, but with a learned model supplying the 'make it a plausible building' step those tools lack).

Core thesis. One can never enumerate every building in a dataset. So rather than scaling data, we teach the model to (i) TRANSFORM any rough input onto the manifold of plausible buildings, and (ii) COMPOSE it from understood architectural elements (windows, doors, roofs, towers). Generality comes from transform + composition, not data size.

2. System pipeline

```
OSM tile / footprint image
| footprint extraction
v per building: (footprint, class, height, style)
[ (1) MASSING PRIOR -- VQVAE + conditional latent diffusion over SDF
  sculpt = SDEdit: encode edit -> partial-noise -> denoise -> clean building SOLID ]
[ (2) ELEMENT COMPOSITION -- part-composer (learned on real labels) decides
  glazing/roof/dome/towers/steps; sdf_detail instantiates windows, DOOR, roof, landmarks ]
| marching cubes . place . compose
v
navigable 3D town -> select a building -> sculpt -> (1) re-snaps, (2) re-details
```

The split is deliberate: (1) supplies coherent MASSING (the overall solid); (2) supplies the BUILDING-NESS (crisp, understood elements). A 64^3 implicit prior excels at the former and is fundamentally limited at the latter, so element crispness is delegated to (2).

3. Method

3.1 Massing prior (1): latent diffusion + SDEdit

A VQVAE encodes a 64^3 signed-distance field to a compact latent; a conditional diffusion model (footprint, class, height, style) is trained in that latent space. Sculpting uses SDEdit: the crude edit is encoded, partially noised, and denoised by the prior - the imposed coarse structure survives while the result snaps onto the manifold of real buildings. A single 'strength' knob trades faithfulness vs realism.

3.2 Element composition (2): part-composer + detail

A conditional diffusion part-composer trained on real BuildingNet part labels (window, wall, roof, dome, tower, stairs) predicts, from a building's massing, which elements it should have and roughly where. A procedural detail module instantiates them as solid, differentiable SDF primitives (recessed window grids, a ground-floor door, gabled/hipped roofs, fused corner towers, a roof-mounted dome).

4. Data

We train on real, watertight building geometry (non-watertight BuildingNet meshes give a broken thin-shell SDF). Source: the Netherlands 3D BAG open LoD2.2 city model; we extract per-building watertight solids and voxelize to 64^3 signed SDFs via a robust generalized-winding-number sign - 11,776 buildings.



Figure 1. Sample of the real 3D BAG training corpus (40 of 11,776). Real gabled/hipped roofs, L-shapes, set-backs and tower forms - watertight by construction.

Clean procedural training corpus — data/recipe_augmentation_v1 (8 styles x 50k)

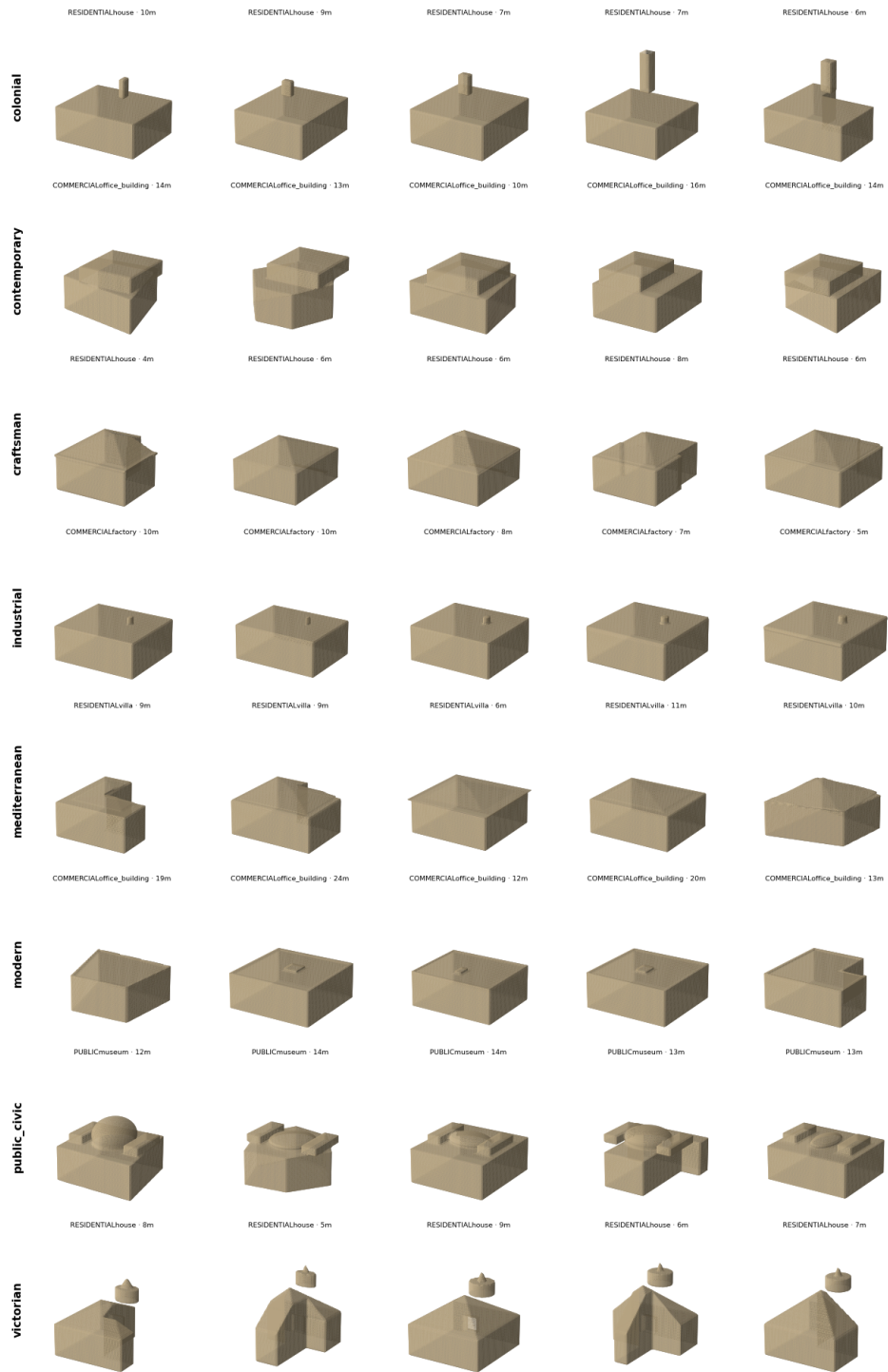


Figure 2. For contrast, our earlier procedural corpus (8 styles): clean but boxy and low-variety - which motivated moving to the real 3D BAG massing above.

5. Results



Figure 3. Sculpting works. Left to right: a real building; a CRUDE user edit (tower spike attached); SDEdit output at increasing strength. Output is a clean, coherent solid - not the speckle that generation-from-noise produces - confirming the editing regime is the right one to deploy.

Composer-driven detail: massing -> learned element placement -> instantiated

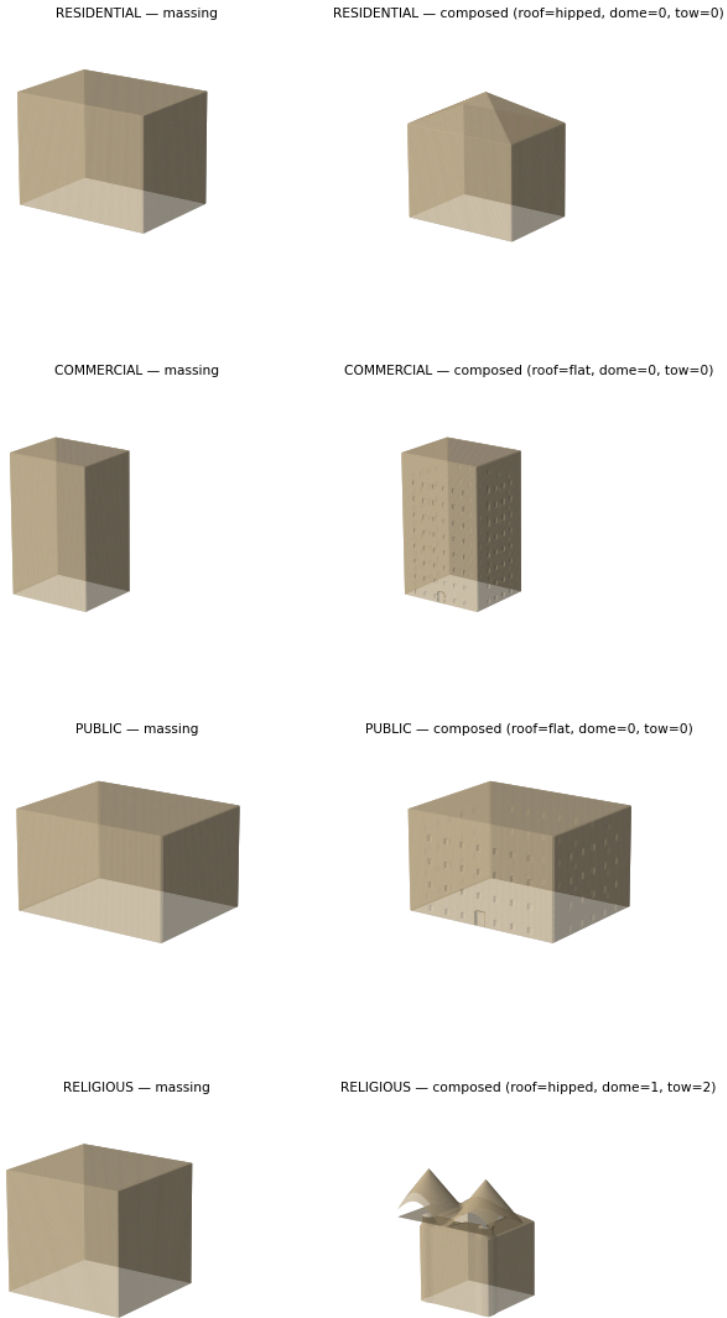


Figure 4. Learned element placement. A boxy massing per class (left) -> the part-composer chooses class-appropriate elements and sdf_detail instantiates them (right). It reproduces real per-class composition from BuildingNet - e.g. religious -> dome + towers; residential -> pitched roofs; commercial/public -> flat.



Figure 5. Training convergence. SDEdit output for the same input across checkpoints (15k/20k/30k). Massing quality improves quickly then largely plateaus - expressiveness instead comes from composition (2) and corpus breadth.

6. Key findings

1. The editing regime beats from-scratch generation. Unconditional generation from pure noise is degenerate (a known metric-vs-inference gap), but SDEdit at partial noise yields clean buildings - so we never sample from noise; we always sculpt/transform.
2. Massing converges fast; data breadth and composition are the real levers, not more iterations (Fig. 5).
3. Part-level coherence requires a part-aware model. Additive (union) composition places elements but cannot guarantee they attach, avoid duplication, and replace prior geometry. We implemented deterministic coherence rules as a stop-gap and specified a learned part-proxy + global-mixing model (SPAGHETTI/SALAD/SPLICE-style) as the principled fix.

7. Status & next steps

(1) SDEdit massing prior (3D BAG)	trained; sculpt output validated clean
Real watertight corpus (3D BAG)	built - 11,776 buildings
(2) part-composer -> detail (+ door)	built and wired into the engine
Part coherence - deterministic rules	done
Part coherence - learned part-mixing	designed (next)
Full OSM -> town -> in-browser sculpt	partial (web demo); final wiring next

Next: (a) route the SDEdit massing + composer through the interactive web sculpt UI to close the OSM->town->sculpt loop; (b) build the learned part-mixing model so element edits re-cohere automatically.

8. Selected references

SDFusion (CVPR 2023) . SDEdit (ICLR 2022) . SPAGHETTI (SIGGRAPH 2022) . SALAD (ICCV 2023) . SPLICE (arXiv:2512.04514, 2025) . StructureNet (SIGGRAPH Asia 2019) . 3D BAG (TU Delft) . BuildingNet . Quilez SDF primitives . libigl (generalized winding number).